

FINAL REVIEW · 23CSE322 FINANCIAL ENGINEERING · TEAM 01

If I Invest *Tomorrow*

A financial decision simulator with an autonomous AI agent

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Indian stocks, ETFs, gold, bonds and cash · data to 1 October 2026 · demo money only

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Introduction

If I Invest Tomorrow tells an investor exactly what to buy tomorrow, how much of each asset, and the probability of reaching their goal — before they invest.

MARKET

26 NSE stocks, 3 index ETFs, gold, government bonds and a liquid fund — 12 years of daily prices.

ENGINE

Machine learning, CAPM, portfolio optimisation, 10,000-path Monte Carlo and stress tests.

AI AGENT

A local language model you talk to in plain English; it plans, simulates and invests with demo money.

| Everything is a simulation with **₹10 lakh of demo money** — no real account or broker is touched.

Objectives

1. Take the investor's **amount, horizon, risk tolerance, target and sectors**
2. **Analyse the market:** returns, risk, correlation, market regime, anomalies
3. **Estimate** expected return and risk with CAPM and machine learning
4. **Build and compare five strategies** on the efficient frontier
5. **Simulate** outcomes and **stress test** against crashes
6. Give **tomorrow's plan** in rupees and shares, and **execute** it through an AI agent

Problem statement

Given an investor's amount, horizon (1, 3 or 5 years), risk tolerance, target return and preferred sectors, decide:

1

What to buy tomorrow

2

How much of each, in rupees and shares

3

The **probability** of reaching the target, and the loss in a crash

Constraint: monthly stock returns are close to unpredictable, so the answer must rest on **risk and probability**, not on a single price forecast.

Literature review

AREA	KEY WORK	WHAT WE USE FROM IT
Portfolio theory	Markowitz (1952); Sharpe (1964)	Efficient frontier, Sharpe ratio, CAPM expected returns
Risk estimation	Ledoit & Wolf (2004); Rockafellar & Uryasev (2000)	Stable covariance matrix; CVaR-minimising crash-resistant portfolio
Market regimes	Hamilton (1989)	Markets switch between calm and crisis states
Simulation	Glasserman (2004)	Monte Carlo for outcome probabilities
ML for returns	Gu, Kelly & Xiu (2020); Chen & Guestrin (2016)	ML adds only small, noisy return signals; tree models as baselines
Anomaly detection	Liu, Ting & Zhou (2008)	Isolation Forest for unusual market days
Robo-advisors	Beketov, Lehmann & Wittke (2018)	Automated allocation is mostly mean–variance based
LLM agents	Yao et al., ReAct (2023); Schick et al., Toolformer (2023)	Language models that call tools instead of guessing answers

Literature gap

WHAT EXISTS

- Prediction tools give a **single price forecast**, with no range of outcomes
- Robo-advisors give an allocation but **no probability of reaching a goal** and no crash test
- ML studies report **weak return predictability**, not a usable plan
- Finance chatbots **invent numbers** and cannot act safely

WHAT WE ADD

- One system: data → models → **plan in rupees and shares**
- **Probabilities**, ranges and **stress tests** for every strategy
- Honest validation: use ML only where it shows skill (volatility)
- An agent **grounded in the engine**, trading only under rules checked in code
- Built for **Indian markets**

Our models vs published research

Our numbers: walk-forward test 2021–2026 on data the models never saw · theirs: as published

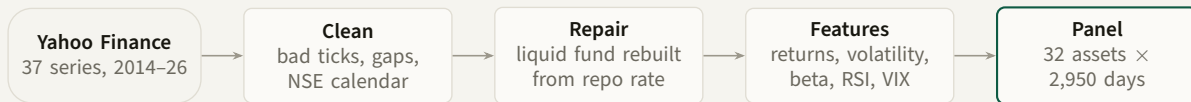
TASK	PUBLISHED RESULT	OUR RESULT	VERDICT
Monthly stock returns (R^2)	Best ML (neural net): 0.40% ; plain regression: −3.46% — Gu, Kelly & Xiu, US stocks	Residual reversal + momentum: +0.09% (same definition); our fitted Ridge: −0.24%	Positive, in the range of top research
Stock direction	LSTM 54.3% — Fischer & Krauss; daily, only the 10 most extreme picks a day, before costs	50.5% monthly on all 32 assets, vs 47.5% for the average	Not like for like; both near 50%
Next-month volatility	Volatility is known to be forecastable from past volatility — Corsi (HAR model)	R^2 0.597 vs 0.50 for the best simple rule, ahead in every year	Works, as the literature predicts
Outcome ranges	No common published benchmark	90% band held 82% of 104 real outcomes	Checked and reported

WHY NOBODY DOES MUCH BETTER

Monthly noise ($\sim 7\%$) swamps the expected return ($\sim 1\%$); professionals trade patterns away; 12 years is only ~ 140 months.

Our edge is not a magic forecast but **honest risk modelling**: the volatility model raises the Sharpe ratio from 0.60 to 0.79.

Methodology 1 of 4: data pipeline



INVESTABLE ASSETS

32

DAILY PRICES

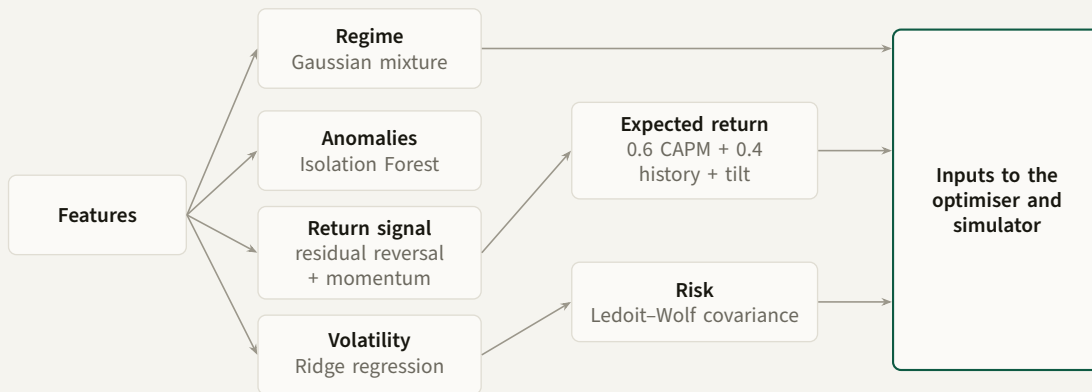
12 yrs

DATA ERRORS FIXED

3

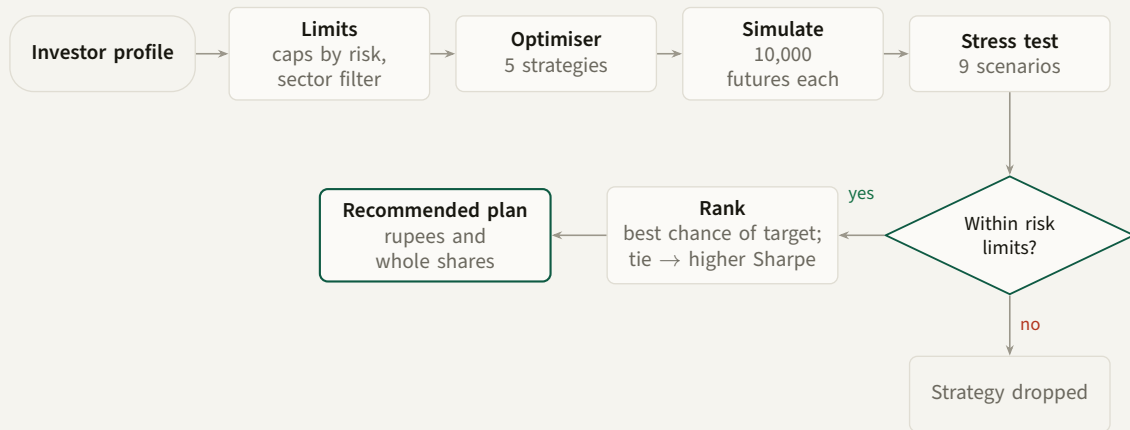
Fixed: wrong-scale prices in 3 ETFs (Dec 2019), erratic gilt-ETF prints, and a liquid-fund price that ignored daily payouts.

Methodology 2 of 4: models



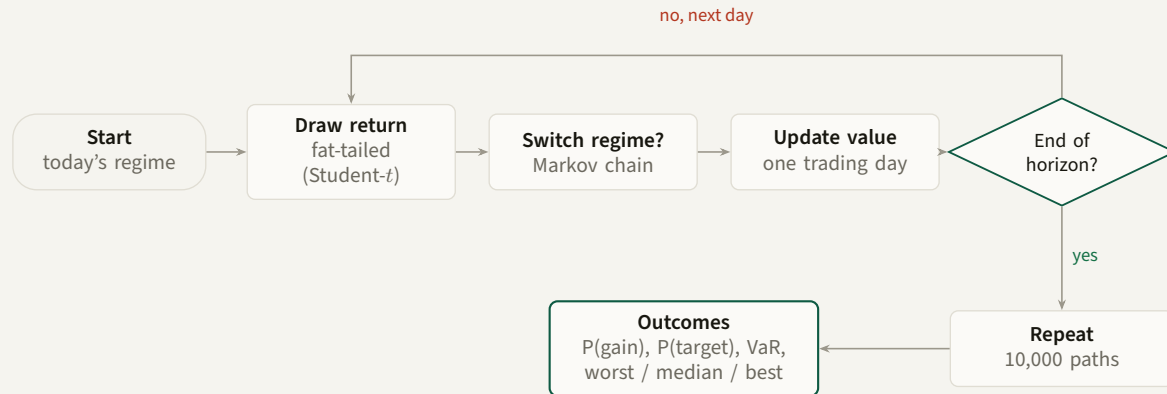
Every model is tested walk-forward on data it never saw (2021–26). Volatility is predictable (R^2 0.60); for returns only a simple residual reversal + momentum signal shows skill (IC 0.05), so it gets only a 23% weight.

Methodology 3 of 4: building and choosing the plan



Five strategies: Max Return · Min Risk · Max Sharpe · Goal-Based · Crash-Resistant (min CVaR)

Methodology 4 of 4: Monte Carlo simulation



Each path also draws its own expected return and volatility level, to include estimation error. A **stress test** starts the paths with a day-0 shock in the bear regime.

Checked against reality: of 104 past forecasts, 82% of outcomes fell inside the 90% band.

Working 1 of 3: four modules on one engine

Planner

Enter a profile; get tomorrow's plan, all five strategies, simulations, stress tests and a backtest.

Simulator

"What if I put ₹X, plus ₹Y a month, into these assets?"
Year-by-year range vs a fixed deposit and NIFTY.

Markets

Today's regime and anomalies, risk and return of every asset, correlations, model test results.

Portfolio

₹10 lakh demo wallet: buy, sell, invest a plan, rebalance, monthly SIPs, health check.

Example: ₹1 lakh + ₹10,000 a month for 10 years in NIFTY, gold and bonds — ₹13 lakh put in, typical result ₹19 lakh, 90% chance of profit.

Working 2 of 3: the AI agent

Qwen3.5-4B runs locally; it chooses the tool and explains — the engine produces every number



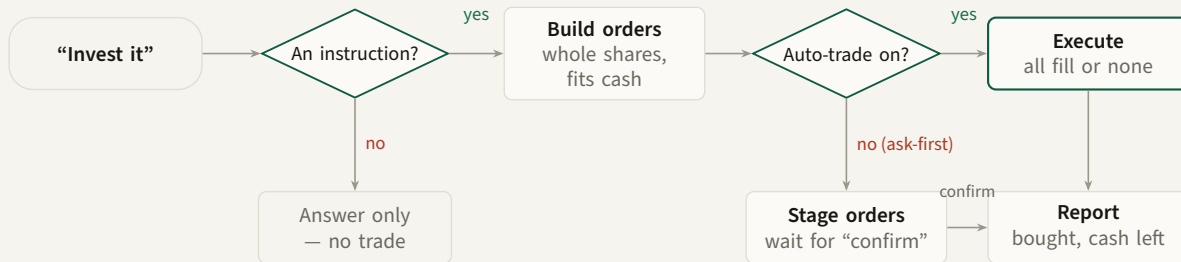
“I have ₹2 lakh for 3 years, medium risk, want 12%” builds and explains tomorrow’s plan

“What’s my chance? Stress test it.” · “Tell me about TCS” runs the simulation, crash tests, stock analysis

“Invest it” · “Sell all my TCS” · “Check my portfolio” trades and manages the demo wallet

23 of 23 test conversations pass. If the model skips a needed tool, the system runs it; the model only explains.

Working 3 of 3: what happens in auto-trade



AUTO-MANAGE RULES

Sell a holding down 15% · trim one that grows too big ·
reduce risk in a crisis · invest idle cash

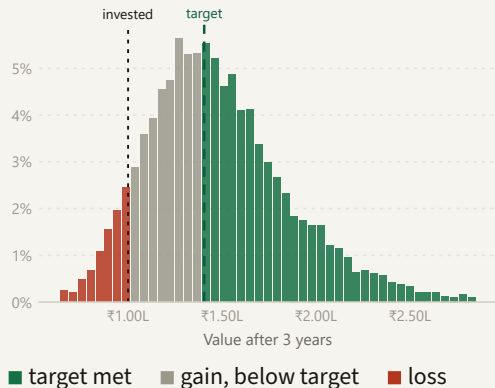
AUTOPILOT

Runs these rules automatically after every market-data update. All checks are in code, not in the language model.

Results

Sample investor: ₹1,00,000 · 3 years · medium risk · 12% target · recommended plan: Goal-Based

RANGE OF OUTCOMES AFTER 3 YEARS



BACKTEST: CHOSEN OCT 2023, HELD TO OCT 2026



EXPECTED RETURN

14.1%

BACKTEST, 3 YRS

+55%

REACH TARGET

54%

NIFTY, 3 YRS

+15%

STRESS TESTS

8 of 9

WORST FALL

—8%

Conclusion

From “*what might happen*” to “*what to do tomorrow*”.

COMPLETE SYSTEM

1

Data, ML, CAPM, optimisation, simulation and stress tests in one pipeline.

STRATEGIES, 1 RULE

5

Compared on the same 10,000 futures and chosen transparently.

AGENT THAT ACTS

1

Plans, simulates, invests and manages — with safety checks in code.

Key finding: next month's returns cannot be predicted reliably, but risk can — so the system's value is in measuring risk, comparing strategies and showing the full range of outcomes.

Future work: taxes and costs, news sentiment, mutual funds, larger local language models.

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Thank *you*

Questions and live demo

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github.com/Tendool/If-I-invest-tomorrow

Simulation with demo money only — not investment advice.